

AMC 10 Style Practice Test

25 problems · 75 minutes · no calculator · choose one answer (A–E) for each problem

Name: _____

Date: _____

1. What is the value of $\frac{2+4+6+\dots+20}{1+3+5+\dots+19}$?

- (A) $\frac{9}{10}$ (B) 1 (C) $\frac{11}{10}$ (D) $\frac{6}{5}$ (E) $\frac{11}{9}$

2. A price first increased by 20%, and then the new price decreased by 25%. How does the final price compare to the original?

- (A) 10% lower (B) 5% lower (C) unchanged (D) 5% higher (E) 10% higher

3. Anna walked the first 3 km at 6 km/h and the next 3 km at 2 km/h. What was her average speed for the whole walk?

- (A) 2.5 km/h (B) 3 km/h (C) 3.5 km/h (D) 4 km/h (E) 4.5 km/h

4. A rectangle has perimeter 34 and area 60. What is the length of its diagonal?

- (A) 10 (B) 11 (C) 12 (D) 13 (E) 14

5. How many two-digit numbers are divisible by 7 but not divisible by 3?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 13

6. The average of five numbers is 12. One number is removed, and the average of the remaining four becomes 13. What number was removed?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 10

7. How many digits does the number $2^{10} \cdot 5^7$ have?

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 17

8. A class has 8 students. A team of 3 must be chosen, but two of the students have quarreled and cannot both be on the team. In how many ways can the team be chosen?

- (A) 40 (B) 44 (C) 48 (D) 50 (E) 56

9. A circle is inscribed in a square with side 8. What is the area of the part of the square lying outside the circle?

- (A) $64 - 8\pi$ (B) $64 - 16\pi$ (C) $64 - 32\pi$ (D) $32 - 16\pi$ (E) $16 - 4\pi$

10. In an arithmetic sequence the first term is 5 and the common difference is 3. Which term of the sequence is the number 302?

- (A) 99th (B) 100th (C) 101st (D) 102nd (E) 103rd

11. A real number x satisfies $x + \frac{1}{x} = 5$. What is $x^3 + \frac{1}{x^3}$?

- (A) 110 (B) 115 (C) 120 (D) 125 (E) 130

12. Two standard dice are rolled. What is the probability that the sum of the numbers shown is a prime number?

- (A) $\frac{1}{3}$ (B) $\frac{2}{5}$ (C) $\frac{5}{12}$ (D) $\frac{1}{2}$ (E) $\frac{7}{12}$

13. A right triangle has legs 6 and 8. What is the radius of its inscribed circle?

- (A) 1.5 (B) 2 (C) 2.4 (D) 2.5 (E) 3

14. What is the least positive integer n such that $n!$ is divisible by 2025?

- (A) 10 (B) 11 (C) 12 (D) 15 (E) 25

15. How many integers n with $1 \leq n \leq 100$ are such that $n^2 + n$ is divisible by 6?

- (A) 33 (B) 50 (C) 66 (D) 67 (E) 83

16. What is the area of a regular hexagon with side 4?

- (A) $16\sqrt{3}$ (B) $24\sqrt{3}$ (C) $32\sqrt{3}$ (D) $48\sqrt{3}$ (E) 96

17. How many four-digit even numbers have all digits distinct and taken from the set $\{1, 2, 3, 4, 5\}$?

- (A) 24 (B) 36 (C) 48 (D) 60 (E) 72

18. What is the sum of all positive divisors of 360, including 1 and 360 itself?

- (A) 1024 (B) 1080 (C) 1120 (D) 1170 (E) 1260

19. What is the sum of all real roots of the equation $||x - 2| - 3| = 1$?

- (A) 0 (B) 4 (C) 6 (D) 8 (E) 10

20. In triangle ABC , point D lies on side BC so that $BD : DC = 1 : 2$. Point E is the midpoint of AD . Line BE meets side AC at point F . What is the ratio $AF : FC$?

- (A) 1 : 2 (B) 2 : 5 (C) 2 : 3 (D) 1 : 4 (E) 1 : 3

21. A sequence is defined by $a_1 = 1$ and $a_{n+1} = 2a_n + 1$ for all $n \geq 1$. What is a_{10} ?

- (A) 511 (B) 512 (C) 1022 (D) 1023 (E) 2047

22. How many integers x satisfy the inequality $|x - 3| + |x + 2| < 9$?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

23. What are the last two digits of 3^{2025} ?

- (A) 01 (B) 07 (C) 27 (D) 43 (E) 83

24. Lattice paths from $(0, 0)$ to $(5, 5)$ consist of unit steps right and up. How many such paths never go above the line $y = x$?

- (A) 14 (B) 32 (C) 42 (D) 120 (E) 252

25. What is the sum of all real numbers x for which $(x^2 - 5x + 5)^{x^2 - 9x + 20} = 1$?

- (A) 9 (B) 10 (C) 13 (D) 14 (E) 15

Answer Key

Parent page: cut off or do not print for the student. Full solutions are available on the test page online.

1	2	3	4	5
C	A	B	D	C
6	7	8	9	10
D	A	D	B	B
11	12	13	14	15
A	C	B	A	C
16	17	18	19	20
B	C	D	D	E
21	22	23	24	25
D	B	D	C	E